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Education and Major Research Training

2023-present	Harvard Medical School; Broad Institute of MIT and Harvard Postdoctoral Research Fellow, Department of Biomedical Informatics Advisor: Dr. Shamil Sunyaev
2015-2022	Massachusetts Institute of Technology PhD in Computational and Systems Biology Advisor: Dr. Manolis Kellis; Significant collaboration with Dr. Li-Huei Tsai Dissertation: "Gene-regulatory circuitry of disease risk and progression"
2011-2015	Princeton University AB in Molecular Biology with certificates in Quantitative and Computational Biology (QCB); <i>Magna Cum Laude</i> ; Outstanding Thesis Award in Molecular Biology; Best Thesis Award for QCB; Sigma Xi Society Advisor: Dr. John Storey Thesis: "Identifying concordance between open chromatin and gene expression"

Funding

08/2025-06/2030	Pathway to Independence Award (K99/R00), NIH/NIA K99 AG095208 <i>Title:</i> Cell-type-specific dysregulation and genetic control of alternative polyadenylation in neurodegeneration Co-mentors: Dr. Shamil Sunyaev, Dr. Rudolph Tanzi, Dr. Vikram Khurana
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Publications

First-author publications:

Google Scholar: [\[IC_JXBgAAAAJ\]](#)

† Co-corresponding, * Co-first authorship

1. Nan J*, **Boix CA**†, Shi S*, Fan X, Ni J, Wang K, Shuai X, Ding K, Wu P, An Y, Sun N, Hou L, Chen K, Huang X, Li C, Akay LA, Louderback K, Nawaid H, Park YP, Fu X, Liu C, Li X, Yin Y, Mo W, Yang Z, Ma W, Hu X, Bennett DA, Kellis M, Tsai LH, Sunyaev S, Li J, Xiong X†. *Single-cell profiling and genetic regulation of alternative polyadenylation in the human brain*. **Nature Genetics**. 2026 Sep. [\[PAPER\]](#) [\[DATA/WEB\]](#)
2. Mathys H*, **Boix CA***, Akay LA*, Xia Z, Davila-Velderrain J, Ng AP, Jiang X, Abdelhady G, Galani K, Mantero J, Band N, James BT, Babu S, Galiana-Melendez F, Louderback K, Prokopenko D, Tanzi RE, Bennett DA, Tsai LH†, Kellis M†. *Single-cell multiregion dissection of Alzheimer's disease*. **Nature**. 2024 Aug;632(8026):858-868. PubMed Central PMCID: PMC11338834. [\[PAPER\]](#) [\[DATA/WEB\]](#) [\[METHOD\]](#)
3. Xiong X*, James BT*, **Boix CA***, Park YP, Galani K, Victor MB, Sun N, Hou L, Ho LL, Mantero J, Scannail AN, Dileep V, Dong W, Mathys H, Bennett DA, Tsai LH†, Kellis M†. *Epigenomic dissection of Alzheimer's disease pinpoints causal variants and reveals epigenome erosion*. **Cell**. 2023 Sep 28;186(20):4422-4437.e21. PubMed Central PMCID: PMC10782612. [\[PAPER\]](#) [\[DATA/WEBSITE\]](#)
4. Dileep V*†, **Boix CA***, Mathys H, Marco A, Welch GM, Meharena HS, Loon A, Jeloka R, Peng Z, Bennett DA, Kellis M†, Tsai LH†. *Neuronal DNA double-strand breaks lead to genome structural variations and 3D genome disruption in neurodegeneration*. **Cell**. 2023 Sep 28;186(20):4404-4421.e20.

PubMed Central PMCID: PMC10697236. [PAPER] [DATA/WEBSITE]

5. Mathys H[†], Peng Z*, **Boix CA***, Victor MB, Leary N, Babu S, Abdelhady G, Jiang X, Ng AP, ..., Bennett DA, Kellis M[†], Tsai LH[†]. *Single-cell atlas reveals correlates of high cognitive function, dementia, and resilience to Alzheimer's disease pathology*. **Cell**. 2023 Sep 28;186(20):4365-4385.e27. PubMed Central PMCID: PMC10601493. [PAPER] [DATA/WEBSITE]
6. Schreiber J[†], **Boix CA***, Wook Lee J, Li H, ..., Epstein C, Strattan JS, Bernstein B, Snyder M, Kellis M, Stafford W, Kundaje A. *The ENCODE Imputation Challenge: a critical assessment of methods for cross-cell type imputation of epigenomic profiles*. **Genome Biology**. 2023 Apr 18;24(1):79. PubMed Central PMCID: PMC10111747. [PAPER]
7. **Boix CA**, James BT, Park YP, Meuleman W, Kellis M. *Regulatory genomic circuitry of human disease loci by integrative epigenomics*. **Nature**. 2021 Feb;590(7845):300-307. PubMed Central PMCID: PMC7875769. [PAPER] [DATA/WEBSITE]

In preparation:

8. Stanton AE*, Kang J*, Blanchard JW*, **Boix CA**, Schroeder ME, Lee Y, Su H, Wang S, Yu E, Emenari A, Peng Z, Agbas E, Cerit O, Park D, Zhang R, Bennett DA, Yin P, Kellis M, Langer R, Boyden E[†], Tsai LH[†]. *Expansion Revealing of Pathology Resolves Nanostructures Associated with Inflammatory Phenotypes in COVID-19 Decedent Human Brain Tissue*. **bioRxiv** (2026) [PAPER]
9. *Kousi M, **Boix CA***, Park YP, Mathys H, Sledzieski S, Peng Z, Bennett DA, Tsai LH, Kellis M. *Single-cell mosaicism analysis reveals cell-type-specific somatic mutational burden in AD*. **bioRxiv** (2022) [PAPER]

Co-Author Publications:

10. Akay LA, Bright A, **Boix CA**, Louderback K, Medrano J, Sun D, Lin H, King O, Welch GM, Nawaid H, Agbas E, Jiang A, Bubnys A, Cheng J, Blanchard J, Tsai LH. *Genome-wide CRISPR screen reveals Wnt signaling defects regulate lipid accumulation in APOE4 oligodendrocytes*. (in press at **Cell Stem Cell**)
11. Liu Z[†], Zhang S*, James BT, Galani K, Mangan RJ, Fass SB, Liang C, Wagle MM, **Boix CA**, Tanigawa Y, Sukwon Y, Sung Y, Xiong X, Sun N, Hou L, Wohlwend M, Qiu M, Han X, Xiong L, Preka E, Huang L, Li WF, Ho LL, Grayson A, Mantero J, Kozlenkov A, Mathys H, Chen T, Dracheva S, Bennett DA, Tsai LH[†], Kellis M[†]. *Single-cell multiregion epigenomic rewiring in Alzheimer's disease progression and cognitive resilience*. **Cell**. 188 (2025) 4980-5002.e29. [PAPER]
12. Castanho I*, Yeganeh PN*, **Boix CA**, Morgan SL, Mathys H, Prokopenko D, White B, Soto LM, Pegoraro G, Shah S, Ploumakis A, Kalavros N, Bennett DA, Lange C, Kim DY, Bertram L, Tsai LH, Kellis M, Tanzi RE, Hide W. *Molecular hallmarks of excitatory and inhibitory neuronal resilience to Alzheimer's disease*. **Molecular Neurodegeneration**. 20 (2025) 103. [PAPER]
13. Von Maydell D, Wright S, Pao PC, Staab C, King O, Spitaleri A, Bonner JM, Liu L, Yu CJ, Chiu CC, Leible D, Scannail AN, Li M, **Boix CA**, Mathys H, Leclerc G, Menchaca GS, Welch G, Graziosi A, Leary, Samaan G, Kellis M, Tsai LH. *ABCA7 variants impact phosphatidylcholine and mitochondria in neurons*. **Nature**. 647 (2025) 462-471. [PAPER]
14. He J, Xiong L, Shi S, Li C, Chen K, Fang Q, Nan J, Ding K, Mao Y, **Boix CA**, Hu X, Kellis M, Li J, Xiong X. *Deep learning prediction of ribosome profiling with Translatomer reveals translational regulation and interprets disease variants*. **Nature Machine Intelligence**. (2024) 1-16. [PAPER]
15. Linna-Kuosmanen S*, Schmauch E*, Galani K, Ojanen J, **Boix CA**, Örd T, ..., Kaikkonen MU, García-Cardena G, Tavi P, Kiviniemi T, Kellis M. *Transcriptomic and spatial dissection of human ex vivo right atrial tissue reveals proinflammatory microvascular changes in ischemic heart disease*. **Cell Rep Med**. 2024 May 21;5(5):101556. PubMed Central PMCID: PMC11148807. [PAPER]
16. Hou L*, Xiong X*, Park Y, **Boix CA**, James B, Sun N, ..., Ardlie K, Kellis, M. *Multitissue H3K27ac*

profiling of GTEx samples links epigenomic variation to disease. Nature Genetics 55.10 (2023): 1665-1676. [PAPER]

17. Sun N, Victor MB, Park YP, Xiong X, Scannail AN, Leary N, Prosper S, Viswanathan S, Luna X, **Boix CA**, James BT, Tanigawa Y, Galani K, Mathys H, Jiang X, Ng AP, Bennett DA, Tsai LH, Kellis M. *Human microglial state dynamics in Alzheimer's disease progression. Cell.* 2023 Sep 28;186(20):4386-4403.e29. PubMed Central PMCID: PMC10644954. [PAPER]
18. Frankish A, et al. **(one of many) GENCODE 2023. Nucleic Acids Research** 51.D1 (2023) D942–D949. [PAPER]
19. Welch GM, **Boix CA**, Schmauch E, Davila-Velderrain J, Victor MB, Dileep V, ..., Kellis M, Tsai LH. *Neurons burdened by DNA double-strand breaks incite microglia activation through antiviral-like signaling in neurodegeneration. Science Advances* 8, eabo4662 (2022). [PAPER]
20. Frankish A, et al. **(one of many) GENCODE 2021. Nucleic Acids Research** 49.D1 (2021) D916–D923. [PAPER]
21. The ENCODE Project Consortium, Moore, JE, Purcaro MJ. et al. **(one of many) Expanded encyclopaedias of DNA elements in the human and mouse genomes. Nature** 583 (2020): 699–710. [PAPER]
22. Leventhal GE, **Boix CA**, Kuechler U, Enke TN, Sliwerska E, Holliger C, and Cordero OX, *Strain-level diversity drives alternative community types in millimetre-scale granular biofilms. Nature Microbiology* 3 (2018): 1295–1303. [PAPER]

Talks

Invited seminars

1. Harvard Medical School - DBMI (2026) - *Concordant pleiotropic effects between pairs of genes reveal molecular mechanisms of disease*
2. University of Chicago (2026) - *Alternative polyadenylation and pleiotropic networks: from genetics to molecular mechanisms*
3. Barcelona Beta Research Center (2024) - *Single-cell multi-region dissection of Alzheimer's disease progression reveals selectively vulnerable neurons*
4. Center for Genomic Regulation (CRG, Barcelona) - PRBB Computational Genomics Seminar Series - (2022) - *Dissecting gene-regulatory circuitry of disease risk and progression in neurodegeneration*
5. CZI/NIH Maps to Mechanisms: Mapping a path forward for ADRD (2021) - *Spatio-temporal multi-omic dissection of Alzheimer's Disease*
6. Program in Quantitative Genomics working group meeting (2020) at Harvard T.H. Chan School of Public Health - *Regulatory genomic circuitry of human disease loci by integrative epigenomics*
7. IHEC and CEEHRC annual meeting on Epigenetics (2019) - *Integrative analysis of 10,000 epigenomic maps across 800 samples for regulatory genomics and disease dissection*

Conference Presentations

8. Human Genetics and Genomics GRS (2025) - *Single-cell landscape and regulation of alternative polyadenylation landscape in the brain*
9. AD/PD (2024) - *Single-cell multi-region dissection of Alzheimer's disease progression reveals selectively vulnerable neurons*
10. American Society of Human Genetics annual meeting (2022) - *A multi-region single-cell molecular atlas of the aged human brain reveals selectively vulnerable neurons in Alzheimer's disease*
11. CSHL Biology of Genomes (2021) - *Single-cell temporal and spatial dissection of Alzheimer's disease*
12. ENCODE Consortium Meeting (December 2019) - *Integrative analysis of 10,000 epigenomic maps across 800 samples for regulatory genomics and disease dissection*
13. ISMB/ECCB annual meeting (2019) - *Leveraging single-cell RNA-seq to infer cell type-specific somatic*

mutations and mosaicism in Alzheimer's disease

14. ROSMAP annual meeting (2019) - *Leveraging single-cell RNA-seq to infer cell type-specific somatic mutations and mosaicism in Alzheimer's disease*
15. ENCODE Consortium Meeting (January 2019) - *Epigenomic imputation across hundreds of ENCODE cell states and types*
16. American Society of Human Genetics annual meeting (2018) - *Leveraging single-cell RNA-seq to infer cell type-specific somatic mutations and mosaicism in AD*

Posters

17. American Society of Human Genetics annual meeting (2026) - *Concordant pleiotropic effects between pairs of genes reveal molecular mechanisms of disease*
18. American Society of Human Genetics annual meeting (2024) - *Regulatory context stratifies genotype-phenotype maps into biological pathways*
19. American Society of Human Genetics annual meeting (2020) - *Regulatory genomic circuitry of human disease loci by integrative epigenomics*
20. American Society of Human Genetics annual meeting (2019) - *Integrative analysis of 10,000 epigenomic maps across 800 samples for regulatory genomics and disease dissection*
21. ENCODE Consortium Meeting, Stanford, CA (2017) - *Learning components of chromatin regulation*

Awards

2024	Junior Faculty Award - AD/PD 2024
2015	Outstanding Thesis Award in Molecular Biology
2015	Best Thesis Award for Quantitative and Computational Biology
2015	Sigma Xi Society

Teaching and Mentoring

Mentorship

2019-present: Graduate mentees: Irtiza Iram, Shivam Gandhi (Harvard); Neil Band (Harvard, M.S.); Benjamin James, Sam Sledzieski (MIT EECS)

Teaching

2016-2021 (Fall) Project mentor for **6.047/6.878: Computational Biology: Genomes, Networks, Evolution**

2017 (Spring) Teaching assistant for **20.440: Analysis of Biological Networks**

2017 (IAP) Co-designed and co-taught **MIT IAP 2017 - Applied Probabilistic Programming & Bayesian Machine Learning**

Other Academic Services and Professional Activities

2019-present **Ad-hoc reviewer for the following journals and conferences:**
Nature Reviews Genetics, Nature Machine Intelligence, Science, Nature Genetics, Nature Methods, RECOMB

2018-present Member of the American Society of Human Genetics
Platform moderator at ASHG 2025

Grant writing: Contributed preliminary data, analysis plans, and helped write science for successfully funded grants including: NIH U01 AG077227, NIH R01 AG058002, NIH U01 NS110453, NIH RF1 AG054012, and NIH R01 MH109978